

WEEK-4

2520090090-Ch.Sai Vihamsh

PART 1 – MySQL Workbench

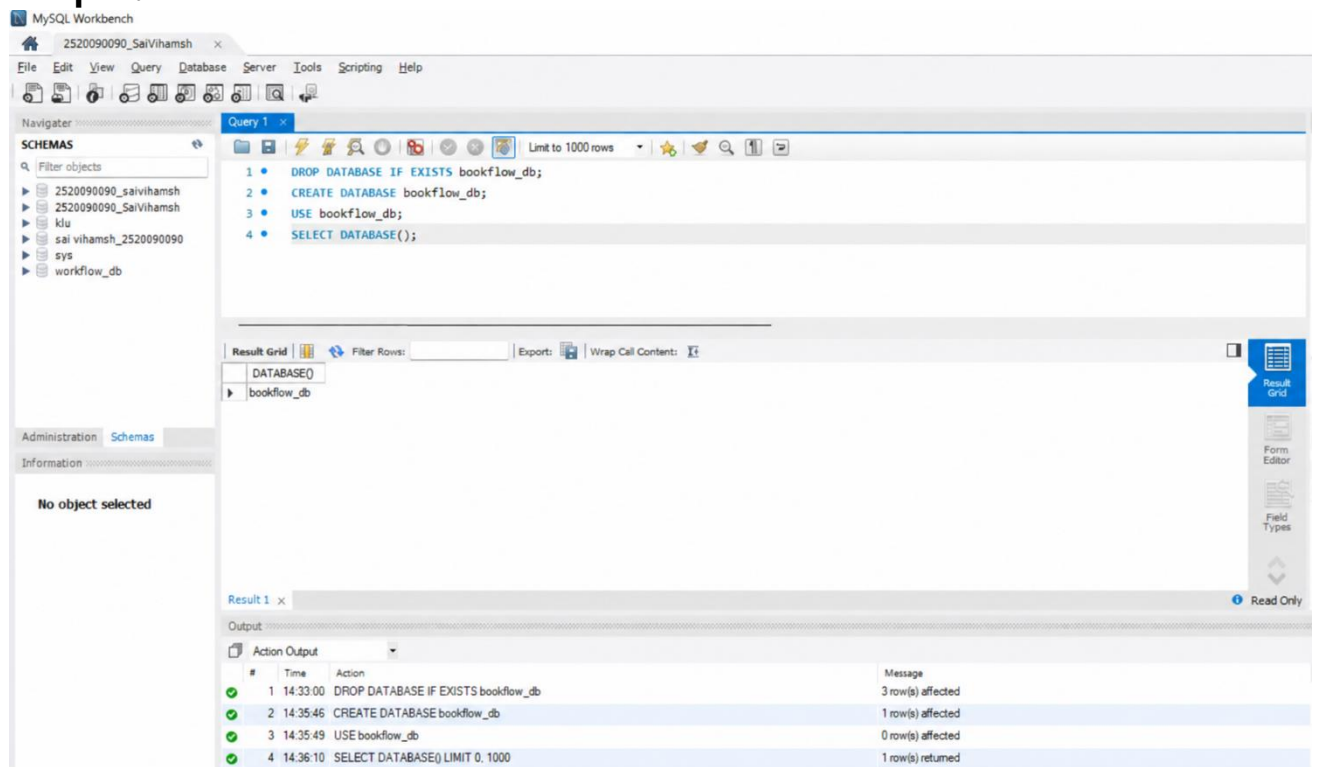
Step 1: Open MySQL Workbench

Open MySQL Workbench and connect to your local MySQL server.

Then click:

File → New Query Tab

Step 2: Create the database



PART 2 – Create Books table

MySQL Workbench

2520090090_SaiVihamsh

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- 2520090090_SaiVihamsh
- 2520090090_SaiVihamsh
- klu
- SaiVihamsh_2520090090
- sys
- workflow_db

Administration Schemas

Information

No object selected

Query 1

```

1 • DROP DATABASE IF EXISTS bookflow_db;
2 • CREATE DATABASE bookflow_db;
3 • USE bookflow_db;
4 • SELECT DATABASE();
5 • CREATE TABLE Books
6 {
7   book_id INT AUTO_INCREMENT PRIMARY KEY,
8   title VARCHAR(200) NOT NULL,
9   author VARCHAR(150) NOT NULL,
10  genre VARCHAR(100),
11  description TEXT,
12  published_year INT
13 };

```

Output

Action Output

#	Time	Action	Message
1	14:33:00	DROP DATABASE IF EXISTS bookflow_db	3 row(s) affected
2	14:35:46	CREATE DATABASE bookflow_db	1 row(s) affected
3	14:35:49	USE bookflow_db	0 row(s) affected
4	14:36:10	SELECT DATABASE() LIMIT 0, 1000	1 row(s) returned
5	14:40:02	CREATE TABLE Books (book_id INT AUTO_INCREMENT PRIMARY KEY, title VARCHAR(200) NOT NU...	0 row(s) affected

MySQL Workbench

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Administration Schemas

Information

No object selected

Query 1

```

8   title VARCHAR(200) NOT NULL,
9   author VARCHAR(150) NOT NULL,
10  genre VARCHAR(100),
11  description TEXT,
12  published_year INT
13 };
14 • DESCRIBE Books;

```

Result Grid

Field	Type	Null	Key	Default	Extra
book_id	int	NO	PRI	<u>NULL</u>	auto_increment
title	varchar(200)	NO		<u>NULL</u>	
author	varchar(150)	NO		<u>NULL</u>	
genre	varchar(100)	YES		<u>NULL</u>	
description	text	YES		<u>NULL</u>	
published_year	int	YES		<u>NULL</u>	

Result 2

Output

Action Output

#	Time	Action	Message
1	14:33:00	DROP DATABASE IF EXISTS bookflow_db	3 row(s) affected
2	14:35:46	CREATE DATABASE bookflow_db	1 row(s) affected
3	14:35:49	USE bookflow_db	0 row(s) affected
4	14:36:10	SELECT DATABASE() LIMIT 0, 1000	1 row(s) returned
5	14:40:02	CREATE TABLE Books (book_id INT AUTO_INCREMENT PRIMARY KEY, title VARCHAR(200) NOT NU...	0 row(s) affected
6	14:41:26	DESCRIBE Books	6 row(s) returned

PART 3 – Insert the books

MySQL Workbench

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Navigator

SCHEMAS

Filter objects

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- sys
- workflow_db

Query 1

```

88 'Twenty Thousand Leagues Under the Seas',
89 'Jules Verne',
90 'Adventure',
91 'A fantastic underwater adventure aboard a mysterious submarine.',
92 1870
93 );
94 SELECT * FROM Books;

```

Result Grid

book_id	title	author	genre	description	published_year

Books 3

Output

Action Output

#	Time	Action	Message
1	14:35:46	CREATE DATABASE bookflow_db	1 row(s) affected
2	14:35:49	USE bookflow_db	0 row(s) affected
3	14:36:10	SELECT DATABASE() LIMIT 0, 1000	1 row(s) returned
4	14:40:02	CREATE TABLE Books (book_id INT AUTO_INCREMENT PRIMARY KEY, title VARCHAR(200) NOT N...	0 row(s) affected
5	14:41:26	DESCRIBE Books	6 row(s) returned
7	14:42:46	SELECT * FROM Books LIMIT 0, 1000	0 row(s) returned

PART 4 – Add embedding column

MySQL Workbench

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Navigator

SCHEMAS

Filter objects

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- klu
- SaiVihamsh_2520090090
- sys
- workflow_db

Query 1

```

85 1997
86 },
87 (
88 'Twenty Thousand Leagues Under the Seas',
89 'Jules Verne',
90 'Adventure',
91 'A fantastic underwater adventure aboard a mysterious submarine.',
92 1870
93 );
94 SELECT * FROM Books;
95 ALTER TABLE Books
96 ADD COLUMN embedding JSON;
97
98
99
100
101
102
103
104

```

Output

Action Output

#	Time	Action	Message
3	14:35:49	USE bookflow_db	0 row(s) affected
4	14:36:10	SELECT DATABASE() LIMIT 0, 1000	1 row(s) returned
5	14:40:02	CREATE TABLE Books (book_id INT AUTO_INCREMENT PRIMARY KEY, title VARCHAR(200) NOT N...	0 row(s) affected
6	14:41:26	DESCRIBE Books	6 row(s) returned
7	14:42:46	SELECT * FROM Books LIMIT 0, 1000	0 row(s) returned
8	14:43:50	ALTER TABLE Books ADD COLUMN embedding JSON	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0

MySQL Workbench

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Administration Schemas

Information

No object selected

Query 1

```

91 'A fantastic underwater adventure aboard a mysterious submarine.',
92 1870
93 );
94 SELECT * FROM Books;
95 ALTER TABLE Books
96 ADD COLUMN embedding JSON;
97 DESCRIBE Books;

```

Result Grid

Field	Type	Null	Key	Default	Extra
book_id	int	NO	PRI		auto_increment
title	varchar(200)	NO			
author	varchar(150)	NO			
genre	varchar(100)	YES			
description	text	YES			
published_year	int	YES			
embedding	json	YES			

Result 4

Output

Action Output

#	Time	Action	Message
4	14:36:10	SELECT DATABASE() LIMIT 0, 1000	1 row(s) returned
5	14:40:02	CREATE TABLE Books (book_id INT AUTO_INCREMENT PRIMARY KEY, title VARCHAR(200) NOT N...	0 row(s) affected
6	14:41:26	DESCRIBE Books	6 row(s) returned
7	14:42:46	SELECT * FROM Books LIMIT 0, 1000	0 row(s) returned
8	14:43:50	ALTER TABLE Books ADD COLUMN embedding JSON	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
9	14:44:34	DESCRIBE Books	7 row(s) returned

Object Info Session

PART 5 – Add mock vectors

MySQL Workbench

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Navigator

SCHEMAS

Filter objects

- 2520090090_SaiVihamsh
- 2520090090_SaiVihamsh
- klu
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- workflow_db

Administration Schemas

Information

No object selected

Query 1

```

131 SET embedding = JSON_ARRAY(0.30, 0.55, 0.95);
132 WHERE book_id = 9;
133
134 UPDATE Books
135 SET embedding = JSON_ARRAY(0.70, 0.90, 0.40)
136 WHERE book_id = 10;
137 SELECT book_id, title, embedding;

```

Result Grid

book_id	title	embedding
9		

Books 3

Output

Action Output

#	Time	Action	Message
5	14:40:02	CREATE TABLE Books (book_id INT AUTO_INCREMENT PRIMARY KEY, title VARCHAR(200) NOT N...	0 row(s) affected
6	14:41:26	DESCRIBE Books	6 row(s) returned
7	14:42:46	SELECT * FROM Books LIMIT 0, 1000	0 row(s) returned
8	14:43:50	ALTER TABLE Books ADD COLUMN embedding JSON	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
9	14:44:34	DESCRIBE Books	7 row(s) returned
10	14:45:53	SELECT book_id, title, embedding FROM Books ORDER BY book_id LIMIT 0, 1000	0 row(s) returned

PART 6 – L2 Similarity Search

MySQL Workbench

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Navigator

SCHEMAS

Filter objects

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- workflow_db

Administration Schemas

Information

No object selected

Query 1

```

163
164 Open a script file in this editor
165
166 );
167 ) AS l2_distance
168
169 FROM Books
170
171 WHERE embedding IS NOT NULL
172
173 ORDER BY l2_distance ASC
174
175 LIMIT 3;
176
177
178
179
180
181
182

```

Output

Action Output

#	Time	Action	Message
✓ 42	14:57:02	UPDATE Books SET embedding = JSON_ARRAY(0.30, 0.75, 0.90) WHERE book_id = 7	0 row(s) affected Rows matched: 0 Changed: 0 Warnings: 0
✓ 43	14:57:02	UPDATE Books SET embedding = JSON_ARRAY(0.88, 0.85, 0.15) WHERE book_id = 8	0 row(s) affected Rows matched: 0 Changed: 0 Warnings: 0
✓ 44	14:57:02	UPDATE Books SET embedding = JSON_ARRAY(0.20, 0.55, 0.95) WHERE book_id = 9	0 row(s) affected Rows matched: 0 Changed: 0 Warnings: 0
✓ 45	14:57:02	UPDATE Books SET embedding = JSON_ARRAY(0.70, 0.90, 0.40) WHERE book_id = 10	0 row(s) affected Rows matched: 0 Changed: 0 Warnings: 0
✓ 46	14:57:09	SELECT book_id, title, embedding FROM Books ORDER BY book_id LIMIT 0, 1000	0 row(s) returned
✓ 47	14:57:50	SELECT book_id, title, author, genre, ROUND(SQRT(POW(CAST(USO...	0 row(s) returned

Object Info Session

PART 7 – Cosine Similarity

MySQL Workbench

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Navigator

SCHEMAS

Filter objects

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- 2520090090_saivihamsh
- klu
- sai vihamsh_2520090090
- sys
- workflow_db

Administration Schemas

Information

No object selected

Query 1

```

221
222
223
224
225 ) AS cosine_similarity
226
227 FROM Books
228
229 WHERE embedding IS NOT NULL
230
231 ORDER BY cosine_similarity DESC
232
233 LIMIT 3;
234
235
236
237
238
239
240

```

Output

Action Output

#	Time	Action	Message
✓ 43	14:57:02	UPDATE Books SET embedding = JSON_ARRAY(0.88, 0.85, 0.15) WHERE book_id = 8	0 row(s) affected Rows matched: 0 Changed: 0 Warnings: 0
✓ 44	14:57:02	UPDATE Books SET embedding = JSON_ARRAY(0.20, 0.55, 0.95) WHERE book_id = 9	0 row(s) affected Rows matched: 0 Changed: 0 Warnings: 0
✓ 45	14:57:02	UPDATE Books SET embedding = JSON_ARRAY(0.70, 0.90, 0.40) WHERE book_id = 10	0 row(s) affected Rows matched: 0 Changed: 0 Warnings: 0
✓ 46	14:57:09	SELECT book_id, title, embedding FROM Books ORDER BY book_id LIMIT 0, 1000	0 row(s) returned
✓ 47	14:57:50	SELECT book_id, title, author, genre, ROUND(SQRT(POW(CAST(USO...	0 row(s) returned
✓ 48	14:59:42	SELECT book_id, title, author, genre, ROUND((CAST(JSON_EXTRACT(embeddin...	0 row(s) returned

Object Info Session